Smart Agriculture AI — Comprehensive Farm Report

Generated: March 13, 2026 at 09:40

1. Soil Analysis

Parameter	Value
Detected Soil Type	Clay
Detection Confidence	69.2%
Detection Method	EfficientNetV2 + SVM

2. Current Soil Conditions (IoT Sensor)

Parameter	Value	Status
Soil Temperature	31.2°C	Normal
Soil Humidity	64.0%	Normal
Soil Moisture	56%	Healthy
Irrigation Status	N/A	

3. Weather Conditions

Parameter	Value
City	Bhopal, India
Outside Temperature	28.2°C
Humidity	53%
Rainfall	0.0 mm/hr
Wind Speed	8.7 km/h

4. Field Details

Total Area: **10,000 m²** (2.47 acres)

5. AI Crop Recommendation

Mode: **Multi-Crop (Zone Allocation)**

Crop	Area (ac)	Alloc %	Yield (kg)	Price/kg	Profit (■)	Rating

Cotton	0.66	26.5%	663	■51.4	■34,096	Good
Rice	0.65	26.4%	1,188	■15.79	■18,752	Good
Wheat	0.58	23.5%	883	■21.25	■18,757	Fair
Lettuce	0.58	23.5%	5,296	■45	■238,320	Fair
TOTAL	**2.47**	**100%**	**8,030**		**■309,925**	

6. Crop Selection Reasoning

Cotton (Score: 0.62):
• [OK] Compatible with Clay soil
• [!] Temperature 31.2°C outside ideal range (21-30°C)
• [OK] Soil moisture 56% ideal (range: 30-60%)
• [i] Current rainfall: 0.0mm/hr
• [!] Higher risk crop (risk factor: 0.6)

Rice (Score: 0.62):
• [OK] Compatible with Clay soil
• [OK] Temperature 31.2°C ideal (range: 20-35°C)
• [!] Soil moisture 56% outside ideal (60-100%)
• [i] Current rainfall: 0.0mm/hr

Wheat (Score: 0.55):
• [OK] Compatible with Clay soil
• [!] Temperature 31.2°C outside ideal range (15-25°C)
• [OK] Soil moisture 56% ideal (range: 40-70%)
• [i] Current rainfall: 0.0mm/hr

Lettuce (Score: 0.55):
• [OK] Compatible with Clay soil
• [!] Temperature 31.2°C outside ideal range (10-20°C)
• [OK] Soil moisture 56% ideal (range: 50-80%)
• [i] Current rainfall: 0.0mm/hr

Report generated by Smart Agriculture AI — IoT + Deep Learning System
Cost savings: Farmers typically spend ■10,000-20,000 on lab soil testing.